

For convergent nozzle, $A_e = A_t = 0.1 \text{ m}^2$

$$\dot{m} = \rho_e u_e A_e = 649.2 \frac{\text{kg}}{\text{s}}$$

$$\tau_{\text{conv}} = \dot{m} u_e + (p_e - p_a) A_e = (0.6771 + 0.5645) \times 10^6 = 1.242 \text{ MN}$$

$\swarrow$ 649.2 $\downarrow$ 1043 $\downarrow$ 5.645 $\times 10^6$ $\swarrow$ 0 $\swarrow$ 0.1

$$C^* = \frac{P_0 A^*}{\dot{m}} = \frac{1.0 \times 10^6 \times 0.1}{649.2} = 1540 \frac{\text{m}}{\text{s}}$$

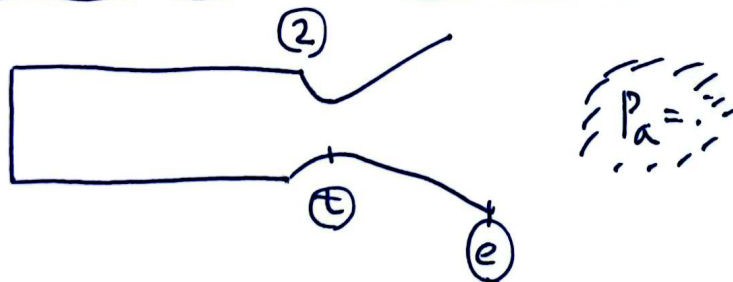
$$C^* = \sqrt{\frac{1}{\gamma} \left(\frac{\gamma+1}{2} \right)^{\frac{\gamma+1}{\gamma-1}}} R T_0 = 1540 \frac{\text{m}}{\text{s}}$$

the same!

$$C_{\tau_{\text{conv}}} = \frac{\tau}{P_0 A^*} = \frac{1.242 \times 10^6}{1.0 \times 10^6 \times 0.1} = 1.242$$

$$C_{\tau_{\text{conv}}} \text{ from } \textcircled{C} = 1.242$$

Case ②



$$A_e = 4.06 \text{ m}^2 \rightarrow \frac{A_e}{A^*} = \frac{4.06}{0.1} = 40.6$$

$$\text{online calc.} \rightarrow M_e = 4.25$$

$$P_e = \left(\frac{P_e}{P_0} \right) \times P_0 = 0.00205 \times 10 \text{ MPa} = 0.0205 \text{ MPa}$$

(P-5)

$$T_e = \frac{T_e}{T_0} \times T_0 = 0.356 \times 3000 = \boxed{1069 \text{ K}}$$

$$u_e = M_e \sqrt{\gamma R T_e} = 2776 \frac{\text{m}}{\text{s}}$$

$$g_e = \frac{P_e}{R T_e} = 0.05759 \frac{\text{kg}}{\text{m}^3}$$

$$\dot{m} = g_e u_e A_e = 649.1 \frac{\text{kg}}{\text{s}} \quad (\text{the same as case (1)})$$

Section 2

$$c^* = \frac{P_0 A^*}{\dot{m}} = 1540 \frac{\text{m}}{\text{s}}$$

$$c^* \text{ from Eq. (c)} = \boxed{1540 \frac{\text{m}}{\text{s}}}$$

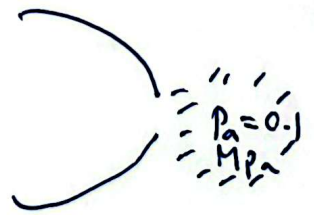
$$\tau = \dot{m} u_e + (p_e - p_a) A_e = (1.802 + 0.083) \times 10^6 = \boxed{1.885 \text{ MN}}$$

$$C_\tau = \frac{\tau}{P_0 A^*} = \frac{1.885}{10 \times 0.1} = 1.885 > C_{\tau \text{ conv}}$$

$$\frac{C_\tau}{C_{\tau \text{ conv}}} = \frac{1.885}{1.242} = 1.52 \quad (> 1 \text{ due to divergence effect})$$

Case (3)

Conv. Nozzle : $P_a = 0.1 \text{ MPa}$, $P_0 = 1 \text{ MPa}$



$$\text{Assume } P_e = P_a = 0.1 \rightarrow \frac{P_0}{P_e} = \frac{10}{0.1} = \boxed{100}$$

$$\frac{P_0}{P_e} = \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{\frac{\gamma}{\gamma-1}} = 100 \rightarrow M_e = 3.4 > 1$$

Wrong!

So $M_e = 1 \Rightarrow$ All the exit condition are the same as Case (1) (P-6)

$$\dot{m} = 649.2 \frac{\text{kg}}{\text{s}}$$

$$p_e = 5.645 \text{ MPa}$$

the same

$$u_e = 1043 \frac{\text{m}}{\text{s}}$$

$$\tau = \dot{m} u_e + (p_e - p_a) A_e = 0.6771 + 0.5545 = \boxed{1.23 \text{ MN}}$$

$\underbrace{\quad}_{0.1 \text{ MPa}}$

$$c^* = 1540 \frac{\text{m}}{\text{s}} \text{ (as before)}, c_{\tau_{\text{conv}}} = 1.23 \text{ (both Eq 7)}$$

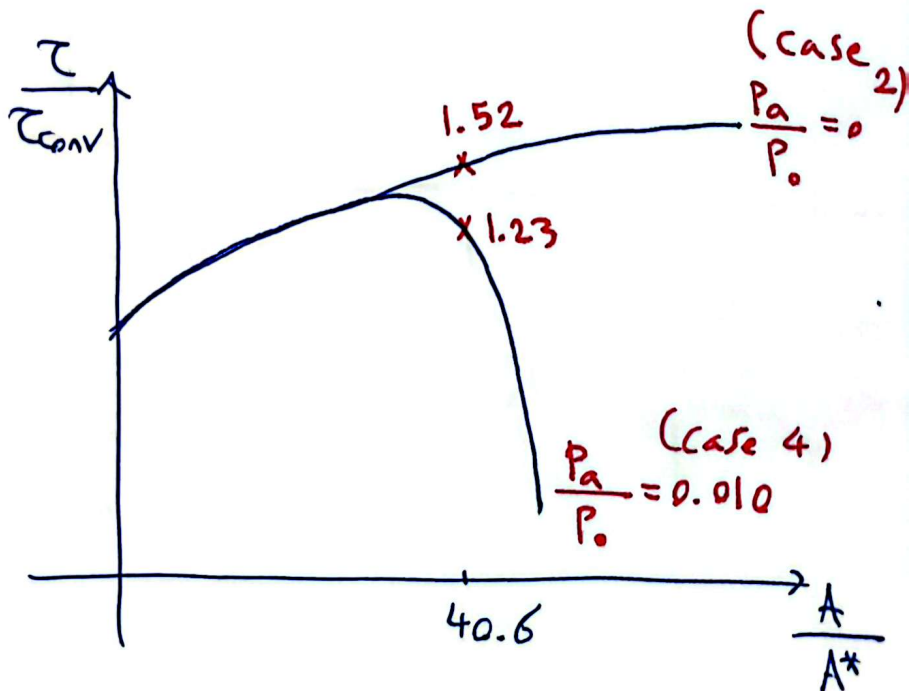
Case (4) $A_e = 4.06 \text{ m}^2$ $\frac{A_e}{A^*} = 40.6$ $\xrightarrow[\text{calc}]{\text{online}}$ $M_e = 4.25$

All the exit conditions ($M_e, u_e, p_e, \dot{m}$) are the same as in case (2) $\rightarrow p_e = 0.020476 \text{ MPa}, c^* = 1540 \frac{\text{m}}{\text{s}}$

$$\tau = \dot{m} u_e + \underbrace{(p_e - p_a) A_e}_{\text{is minus}} = 1.802 - 0.323 = \boxed{1.479 \text{ MN}}$$

$$c_{\tau} = \frac{\tau}{p_0 A^*} = \frac{1.479}{10 \times 0.1} = \boxed{1.479} \text{ (from both Eq 7)}$$

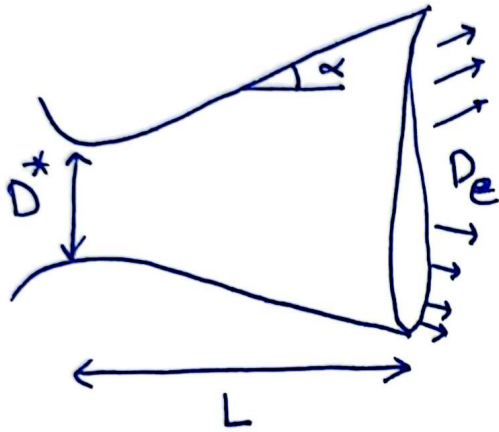
$$\frac{c_{\tau}}{c_{\tau_{\text{conv}}}} = \frac{1.479}{1.23} = 1.20$$



Nozzle shapes

Conical, perfect, maximum thrust ($\equiv$ minimum length),
plug

Conical Nozzle (see Fig. 11.4)



Exit flow is not axial (flow angularity)
Some thrust is generated in the transverse direction.

$$\frac{D_e}{2} = \frac{D^*}{2} + L \tan \alpha$$

$$\Rightarrow D_e = D^* + 2L \tan \alpha$$

$$\frac{D_e}{D^*} = \left(1 + \frac{2L \tan \alpha}{D^*}\right), \quad \frac{A_e}{A^*} = \left(\frac{D_e}{D^*}\right)^2 = \left(1 + \frac{2L \tan \alpha}{D^*}\right)^2$$

$$\Rightarrow L = \frac{D^*}{2 \tan \alpha} \left[\sqrt{\frac{A_e}{A^*}} - 1 \right]$$

Divergent loss for the conical nozzle (λ)

$$\lambda = \frac{1 + \cos \alpha}{2}$$

$$\text{So } \zeta_{\text{conical nozzle}} = \lambda \zeta_{\text{Eq}} = \lambda \dot{m} u_{\text{eq}} = \dot{m} u_{\text{eq}} \frac{1 + \cos \alpha}{2}$$

Advantage: Simple

Disadvantage: Exit flow is not completely axial, which introduces divergent loss (λ) (P-3)

$$\alpha = 0$$

This has minimum thrust angularity
but can not increase M_e !

$$\lambda = \frac{1 + \cos \alpha}{2} = 1$$

(best)

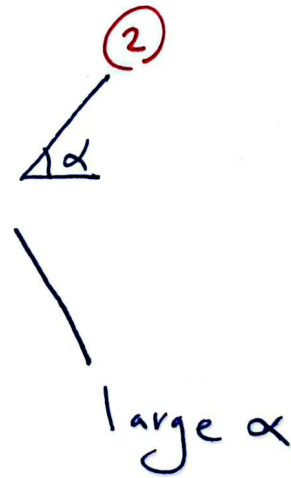
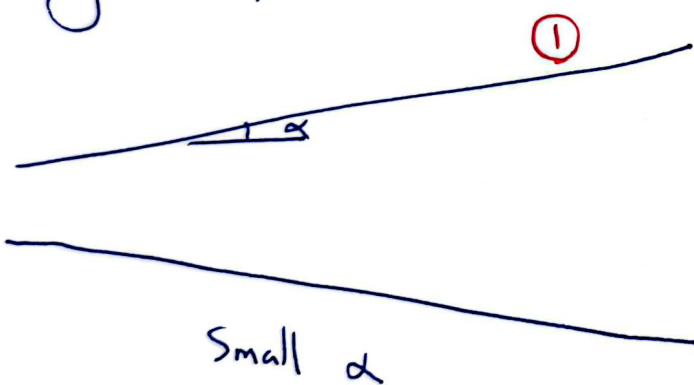
$$\alpha = 90^\circ$$

$$\lambda = \frac{1 + \cos \alpha}{2} = \frac{1}{2} \text{ worst case!}$$

$$\alpha = 90^\circ$$

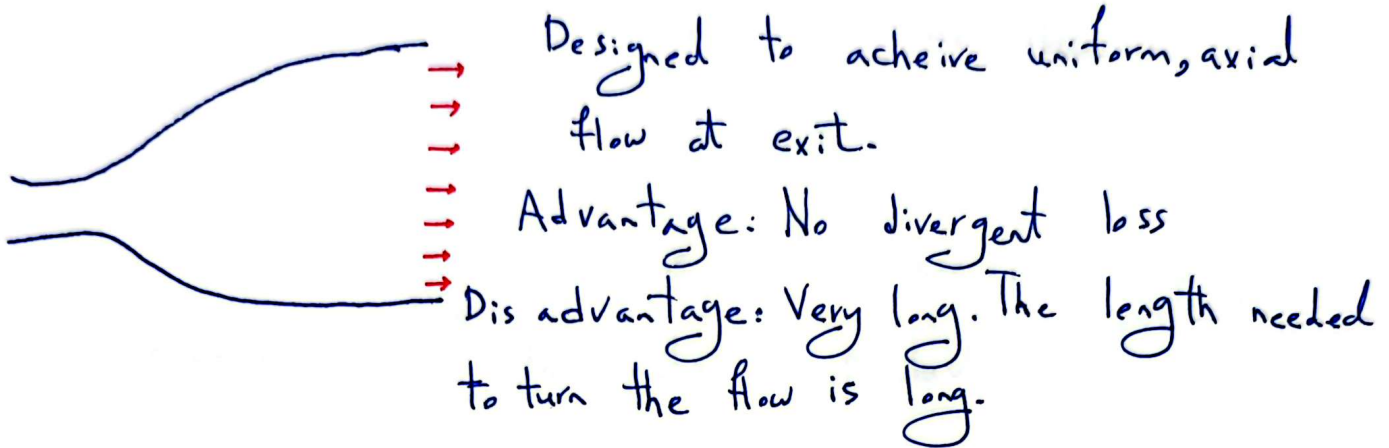
50% thrust reduction

Comment: λ can be reduced by decreasing nozzle half angle (α)



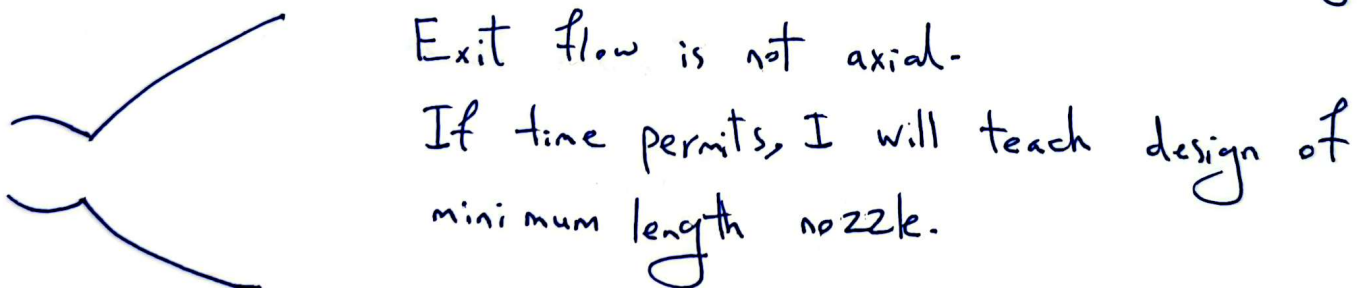
Decreasing α results in larger nozzle for a given area-ratio ($\frac{A_e}{A^*}$). We need a trade-off between ① and ②

Perfect Nozzle

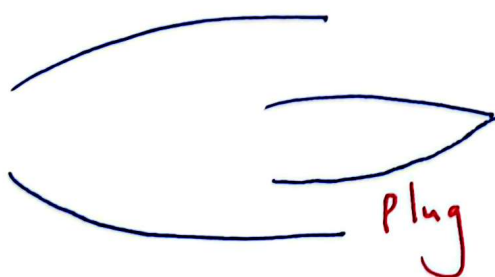


"Maximum thrust or minimum length nozzle"

Designed to give maximum thrust for a specified length.



plug Nozzle (supersonic aircrafts)



~~plug~~ plug: we have oblique shocks rather than normal shocks